

What Is Recoverable?

An Error-Budget Approach to County-Transfer Outage Forecasting

INFORMS 2026 DMS Data Challenge

Abstract

We forecast hourly county Outage Severity Index (OSI) at four horizons for 63 counties never seen in training, through a two-wave March 2026 wind event. Rather than searching model space, we first measured *how the residual error decomposes*, because that determines where attention is worth spending. The diagnosis — computed before the later experiments, from cached out-of-fold predictions and without retraining — puts the error at roughly 45% county *scale*, 37% trajectory *shape* and 17% *timing*, with no dominant mode, and finds that whatever a model still gets wrong about a county's overall magnitude is not recoverable from any legal feature we could construct (out-of-fold $R^2 \approx 0$; -0.09 with 219 features and boosted trees). That diagnosis was turned into falsifiable predictions for eight *intervention families*, and 210 controlled runs plus a further 28 paired contrasts were used to test them. The resulting forecaster is an equal-weight 50/50 blend of two halves — eleven diverse models sharing one half, and a single high-variance-reduction model holding the other — with **no fitted combination parameters**, scoring **0.00838** (mean of the four horizon RMSEs with the configuration choice itself nested; a reporting summary, not the official ranking statistic) against **0.01632** for all-zeros and **0.01110** for decayed persistence. The full ledger, the code, and an interactive dashboard over all 210 runs are in the repository named in §6.

The ensemble's most valuable members are its *worst* models: the three stock-flow models are the weakest of the eleven and the three least correlated with the rest, and no substitute survived a replacement test. Separately, the largest single improvement we made to a member made the ensemble *significantly worse* ($P=0.011$), and ten further variance-reduction runs reproduced that pattern. Both follow from one property of a 239-county sample: a member's marginal value to the ensemble is **not determined by its solo RMSE**, and residual complementarity can dominate it.

1 Task, protocol, and diagnosis

Task. Each county has a 216-hour event window. Hours 0–71 are observed for every county, test counties included; outage variables are withheld from hour 72 onward. The submission is indexed by 144 origins per county (hours 72–215) and four horizons, so the scored targets span hours 73–215 — 143 distinct hours, since hour 72 is an origin but never a target. Weather is given for all 216 hours, so the exogenous drivers are known across the forecast window: the horizon is twice the context and the covariates carry the dominant signal. Training and test counties are disjoint (239 / 63), so this is *cross-sectional* transfer — the model sees a test county's first 72 hours but never its storm outcome, and no post-origin outage value is available at inference. 42.4% of scored cells are exactly zero and the target has skew 8.8.

Metric. RMSE is computed separately for each of four horizons ($t+1$, 6, 24, 48 h) over 9009/8694/7560/6048 valid cells, and teams are ranked *within* each horizon with the four ranks averaged. There is no official scalar; the mean of the four RM-

SEs used throughout this report is a reporting convenience, and we verify that every design decision improves all four horizons rather than only the mean.

Protocol. Five-fold county-holdout cross-validation, stratified by state $\times$ severity. All outage-derived columns are set to NaN after hour 71 *before* feature construction, so validation and deployment share one code path; overwriting every post-origin outage value with garbage changes predictions by exactly 0.0. Anything learned across folds — a scalar, a member set, a rule — is fit on four folds and applied to the fifth, and we report the nested figure wherever a choice was made by looking at scores. Every P and interval below comes from one procedure: a *paired county bootstrap* that resamples the 239 counties with replacement, applies the same resample to both models, recomputes the per-horizon mean RMSE for each, and reports the fraction of resamples in which the candidate wins together with a percentile interval on the difference.

Diagnosis. Decomposing the residual of an early ensemble by substituting each of a county's true scale, shape and timing in turn gives roughly 45/37/17, with no dominant mode. Two further measurements sharpen it. First, the per-county residual multiplier — what a model still gets wrong about a county's overall magnitude — is not predictable out of fold from any legal feature set we built, including 219 aggregated features with boosted trees ($R^2 = -0.09$). Second, the error is spatially concentrated: 14 of 239 counties carry 65% of it, and 82% of squared error is *under*-prediction of outages that did occur. Exact zeros, despite being 42.4% of cells, carry 0.7%.

This is an empirical statement about what our features and learners recover, not a proof of irreducibility. Read that way it still generates sharp predictions: interventions that add information or change the function class should return little, because the missing quantity is not in the features and is not a function-class artefact; interventions that reduce estimator variance or make models fail *differently* should return more, because 191 sequences per fold is a small-sample regime. The rest of the paper tests those predictions.

2 The intervention program

An experiment is classified by the *contrast* it was run to test, not by the properties of the model it produced. A shipped member may be a bi-GRU with a dual objective and snapshot averaging; the run that admitted it changed exactly one of those against a control. Eight mutually exclusive families exhaust our experimental ledger:

F1 signal · **F2** mechanism · **F3** target · **F4** exposure · **F5** estimation · **F6** adaptation · **F7** combination · **F8** uncertainty

plus two book-keeping codes: **P** for procedures whose object is choosing or evaluating rather than predicting, and **X** for

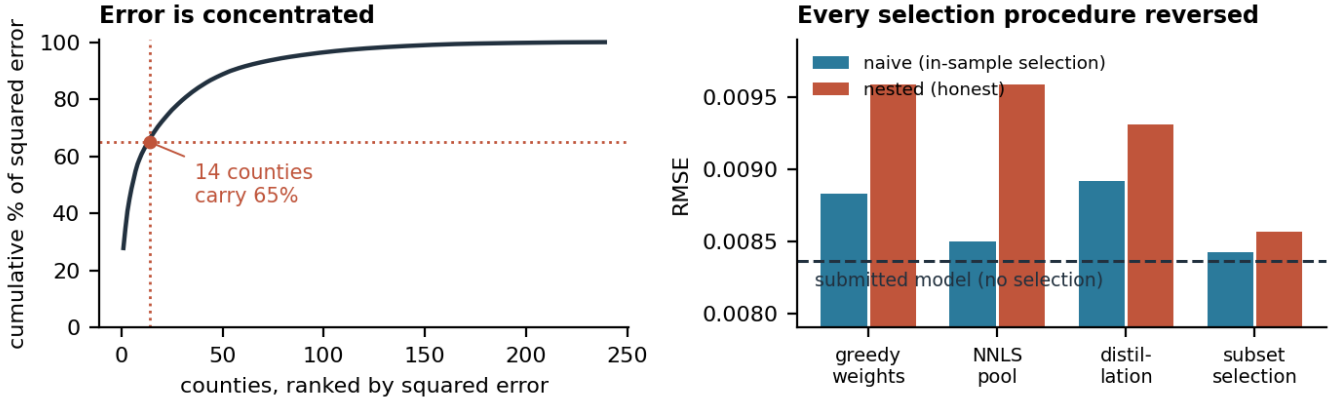


Figure 1: Left: squared error against counties ranked by their contribution — 14 counties carry 65%. Right: four selection procedures, scored naively and with the selection itself nested inside the folds. Every one reverses.

runs that deliberately changed more than one stage and so cannot serve as clean evidence for either. Table 1 is the record; `family_ledger.py` regenerates it from the run log. Four families — F1, F2, F5 and F6 — had the *direction* of their outcome written down before the corresponding runs existed; the rest complete the ledger.

F2, in detail, because it is the family most readers expect to pay. Nineteen architectures were run on this protocol, each adapted to consume the known-future weather so that none was tested where it is weakest. The 2023–2026 exogenous-covariate family lands at 0.0095–0.0120 (PatchTST 0.0095, TSMixer 0.0098, iTransformer 0.0102, TimeXer 0.0108, XLinear 0.0111, DLinear 0.0120) against 0.0088–0.0090 for the recurrent members. Only one architecture outside the recurrent family earned a seat, and not for its accuracy.

Foundation models, measured rather than dismissed. The obvious modern move is a pretrained forecaster, and the obvious objection — that these models assume univariate input with no known future — stopped being true with the covariate-aware generation, so we ran one. **Chronos-2** (120M) scores 0.0213 zero-shot given our 23 weather covariates and 0.0201 without them: both *worse than predicting zero everywhere* (0.0163), and the covariates make it worse rather than better. The failure is diagnosable — it places non-zero mass almost everywhere, 3.2% exact zeros against 42.4% in the truth, so it cannot represent a target silent on nearly half its cells. Four tabular foundation models (TabPFN, TabDPT, TabICLv2, Mitra) were prepared as drop-in learners for the tabular member on its exact rows; two are gated behind a licence or a GPU, and the rest were stopped once the sequence result made the outcome clear. None is shippable regardless: a checkpoint pretrained on outside corpora is external data of a kind the self-contained-code requirement does not admit. We report the number because “foundation models do not transfer here” is worth more as a measurement than as an assertion.

F3, in detail, because skew dominates the target. With 42.4% zeros and skew 8.8, squared error on the raw scale is set by a handful of county-hours. Fitting on $\sqrt{\text{OSI}}$ and inverting improves the same gradient-boosted model from 0.01028 to 0.00967; $\log_{1p}$ over-corrects (0.00996). But squaring a $\sqrt{-}$

scale fit back is biased by Jensen’s inequality, and RMSE rewards the conditional mean: the tabular member predicts 0.841 of the truth on average. The members that matter therefore use a *dual* objective, half the error on each scale, which buys the variance stabilisation without the back-transform bias. Removing the bias entirely from the tabular member — Duan’s smearing estimator, ratio $0.841 \rightarrow 1.01$ — improves that member alone and makes the ensemble *worse*; the partial rescale that all five folds adopt leaves it shrunk. The blend wants that member biased low, which is the shrinkage result of §1 arriving from a second direction.

The same family covers the probabilistic objectives. A conditional normalizing flow trained by maximum likelihood, with its sample mean as the point forecast, reaches 0.0099 — respectable, and not admitted, because RMSE rewards the conditional mean and an MSE-trained network estimates it directly rather than through sampling. Tweedie and quantile heads behave the same way: better models of the distribution, worse answers to the question asked.

Two-stage and hurdle models, tested three ways and rejected. The obvious reading of 42% zeros is a two-part model, and it is what the operational outage literature reaches for [15]. A perfect zero classifier would move the score by 0.00003, because those cells carry 0.7% of the error. Measured, as a target factorisation: $P(y>0) \times E[y | y>0]$, fitted separately on the tabular member’s rows, is worse on all four horizons than the single-stage model, because training the magnitude head on positives alone discards what the zeros say about magnitude. The corresponding *post-hoc* versions — a boosted classifier gating the finished ensemble, and a random-forest gate splitting counties before choosing a configuration per group — are F6 and fail there: the county gate is worse by 0.0004, and still worse by 0.00003 when its gate is replaced by the true grouping. The zeros are also not a phase: counties show a median of four separate zero runs, and 178 of 239 are already non-zero at the origin, so there is no onset to predict.

3 The forecaster the evidence leaves

Three families shaped the final model: F4 created alternative error bases, F5 reduced estimator variance where that gain out-

	family	scored runs	blend contrasts	diagnostic implication	what the contrasts showed
F1	signal	45	—	the residual is unrecoverable from the feature sets tested; with 191 sequences/fold, extra channels should have small or negative marginal value in high-capacity models	external statics and sub-county weather help the tabular member; the same channels <i>lose</i> in 3/3 sequence models, which is the capacity asymmetry the hypothesis predicts; retrieved analogs fail
F2	mechanism	67	—	error is spread across scale, shape and timing, so no function-class substitution should unlock a step change	5 of 6 architecture families land at or behind a plain GRU; Chronos-2 zero-shot scores worse than predicting zero; a mediocre patch transformer is retained only because it fails differently
F3	target	27	—	42.4% zeros are cheap (0.7% of squared error); skew is expensive	$\sqrt{\cdot}$ -target beats raw by 6% and beats log1p; the dual raw/ $\sqrt{\cdot}$ objective and metric-aligned hour weights transfer; Tweedie, quantile, hurdle and a likelihood-trained flow all lose on RMSE
F4	exposure	27	—	robust perturbation can manufacture distinct error bases	storm jitter and mixup survive as distinct bases; multi-origin and dropout variants do not
F5	estimation	23	—	191 sequences/fold implies variance-limited estimators	seed and snapshot averaging improve nearly every model <i>alone</i> ; ten of ten fail to improve the blend
F6	adaptation	7	12	the per-county residual has out-of-fold $R^2 \approx 0$, so post-hoc correction should fail	0 of 12 corrections help: isotonic, power-law, per-hour curves, severity multipliers, and both the zero gate and the county gate. Only a mechanism-backed quiet-tail scalar survives
F7	combination	4	14	with no dominant error mode, complementary residuals should matter	fitted weights, NNLS, distillation and subset selection all reverse under nesting; a parameter-free amplitude $\times$ shape rule wins
F8	uncertainty	1	2	orthogonal to the point metric	split-conformal intervals reach 0.900 coverage at every horizon and leave RMSE unchanged
P	protocol	4	—	—	Ray and Optuna searches; tuned configurations do not beat the untuned control
X	compound	5	—	—	changed two stages at once; kept in the ledger, excluded from family evidence
total		210	28		

Table 1: The experimental ledger; nine further attempts crashed on a code fault without producing output and are excluded. *Scored runs* have their own out-of-fold score; *blend contrasts* are paired tests against the shipped ensemble, which is the only way a combination or correction can be evaluated. F1, F2, F5 and F6 had their direction recorded before those runs existed.

weighed the diversity it costs, and F7 exploited what complementarity remained. What survives is an equal-weight blend of two objects,

$$\hat{y} = \frac{1}{2} \mathcal{C}[f_1, \dots, f_{11}] + \frac{1}{2} A,$$

followed by the quiet-tail scalar. Twelve trained predictors go into the forecast, in two unequal halves.

The **bracket** $\mathcal{C}[f_1, \dots, f_{11}]$ is eleven diverse models sharing half the weight, so each carries 1/22 of the forecast. $\mathcal{C}$ combines them by *amplitude and shape separately* — the mean of their per-county totals times the mean of their normalised trajectories. It is parameter-free, equals the plain mean when the members agree on timing, and preserves peak height when they do not, because averaging in raw space spends amplitude

to hedge timing disagreement.

The **anchor** A is one model holding the other half by itself. It is the same bi-directional GRU as two of the bracket members, but estimated 25 seeds $\times$ 4 cosine-restart snapshots — 100 networks averaged — which makes it the most accurate single object in the project at 0.00862 and by far the lowest-variance. It is called the anchor because that is its job: it holds the forecast steady while the bracket supplies disagreement. It sits outside the bracket rather than inside it for a measured reason. Placed as a twelfth member it would receive 1/12 of the weight, and a paired county bootstrap says that under-weights it — it is as good alone as the whole eleven-member bracket, and the 50/50 split is the only fixed weighting that says so. That split is *not* fitted: 1/2 and 1/11 are the two round num-

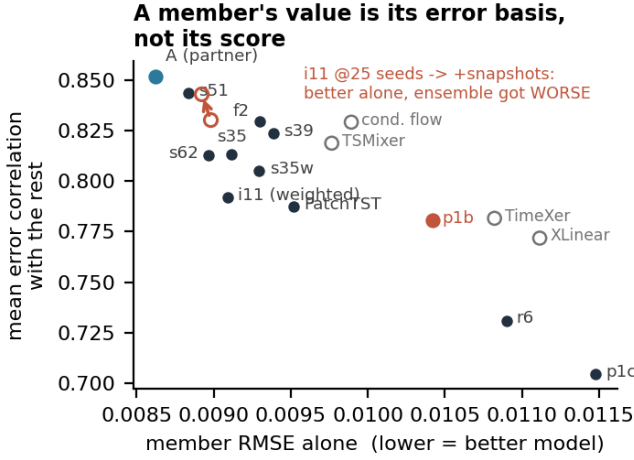


Figure 2: Each object by how good it is alone (horizontal) against how much its errors resemble the rest (vertical). The useful direction is *down*, not left. The three stock-flow models are the worst in the set and the only ones far from the crowd. The arrow shows a member being genuinely improved: it moves left and also up, and the ensemble got worse.

bers the structure implies, and §3 shows what happened every time we tried to fit them instead.

member	mechanism / admitting contrast	alone	corr
A	recurrent; F5 schedule diversity	0.00862	—
x9	recurrent; F5 snapshots	0.00884	0.84
s62	recurrent; F3 hour weights	0.00897	0.81
i11w	recurrent; F4 storm jitter	0.00909	0.79
x4	recurrent; F4 mixup	0.00912	0.82
s35w	recurrent; F3 on the mixup base	0.00930	0.80
f2	trees; F1 external statics	0.00930	0.83
s39	attention; F1 other counties	0.00939	0.82
ts2c	patch transformer; F2	0.00952	0.79
p1b	stock-flow; F2	0.01042	0.78
r6corr	stock-flow; F6 flow scalars	0.01090	0.73
p1c	stock-flow; F4 teacher-forced	0.01148	0.70

Table 2: The anchor *A* — one model holding half the forecast — and the eleven bracket members sharing the other half, each labelled by the family of the contrast that admitted it. Shipped members inherit earlier accepted interventions; the label is the *provenance*, not a claim that they differ in one respect only.

The worst member is the one to keep. p1b is barely better than decayed persistence and ranked first of 28 candidates for its seat; every attempt to replace it made the blend worse at all four horizons. Two further stock-flow variants were admitted later by an add-one sweep over every cached object, and they are the two least correlated members in the set. Doubling the existing physics weight does *not* help, so these are new bases rather than more of the same one.

Improving a member can cost more than it buys. Snapshot-ensembling one member improved it from 0.00955 to 0.00892 — its best score ever — and made the ensemble significantly worse ($P=0.011$), because the technique drove its errors onto *A*’s, and *A* is itself a snapshot ensemble. Re-running

all ten remaining members at 25 seeds turned that anecdote into a controlled sweep: six of the ten improved *alone* — one by 0.00031 — two got worse because they already average internally, and the two tree-based members were unchanged because their seed only moves subsampling. Not one was adopted into the blend on a single fold, and all ten together scored 0.00848 against 0.00835. Variance reduction is the family with the best solo returns and the worst ensemble returns, and the two facts have the same cause.

Nothing in the combination is fitted. Every weight above is fixed in advance. Fitting the full simplex buys 0.000007, while greedy weights, NNLS pooling, distillation and exhaustive subset selection all reverse under nesting (Fig. 1, right) — five folds returned five different answers each time. Exactly two fitted scalars survive anywhere in the model, both nested and both mechanism-backed: the quiet-tail factor, and the tabular rescale of §2.

4 Results and uncertainty

model	$t+1$	$t+6$	$t+24$	$t+48$	mean
all-zeros	23.49	19.82	12.09	9.87	16.32
decayed persistence	12.79	11.92	10.19	9.49	11.10
best single model (<i>A</i>)	10.16	9.23	7.90	7.20	8.62
submitted ensemble	9.94	9.07	7.57	6.83	8.35

RMSE $\times 10^3$. The four horizon columns are what the organisers score; *mean* is our reporting summary. The horizons are not comparable to each other — $t+48$ scores only hours 120–215, after the second wave, where the target is smaller and more often zero — but they are comparable across teams, which is how the ranking works.

The ensemble improves on the strongest naive baseline by 25% and removes 73% of the target’s squared magnitude, against 54% for persistence. It is better than the eight-member predecessor at all four horizons. Two figures should be distinguished: 0.00835 is the out-of-fold score of the fixed final configuration, and **0.00838** is the estimate when the choice among the ten configurations searched is itself nested inside the folds. The second is the honest one and is what we quote wherever a single number carries the claim; the 0.00003 between them is what the selection is worth, and reporting only the first is the mistake the right-hand panel of Fig. 1 is about. Figure 3 shows forecasts for four held-out counties.

Where the blend earns its keep. Hours 72–95 carry 51% of all squared error, and there the eleven-member blend scores 0.01777 against 0.01778 for *A* alone: it contributes nothing. From hour 96 onward it beats decayed persistence by 20–39%. The reason is visible in the same window: on the 24 counties holding 84.5% of early error, the model’s correlation with truth is 0.887 and its bias is 0.887 — the shape is right and the level is 11% low. Every member makes that same error, so there is no disagreement for a blend to exploit. Diversity pays only where members genuinely differ. This also explains why severity-targeted corrections fail: severity itself *is* predictable — the ensemble’s own county totals rank it at AUC 0.985 — but the counties we get *wrong* overlap the most severe ones only 6 of 13, and a multiplier keyed on the best available severity ranking is worse on all four horizons out of fold.

Forecasts for counties never seen in training (shaded = 72 h of observed history; all later hours predicted)

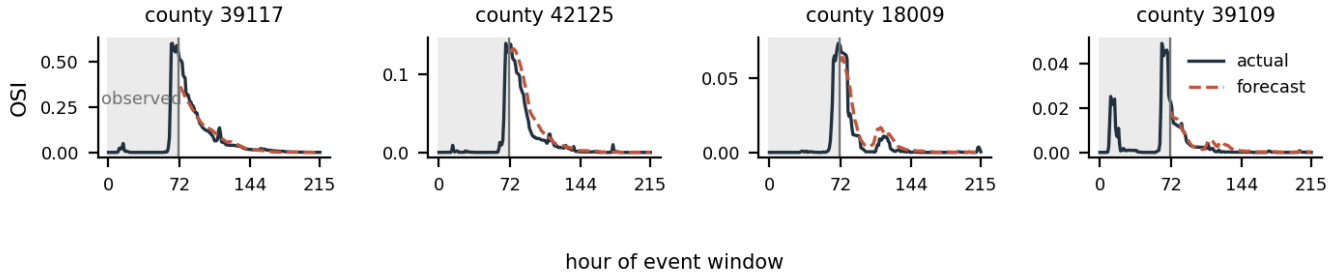


Figure 3: Forecasts for four counties never seen in training, spanning the severity range. Shaded: the 72 observed hours. Everything after is predicted.

Conformal intervals. A point forecast with no statement of its own uncertainty is a weak deliverable for an operational task, so we calibrated split-conformal intervals on four folds and checked them on the fifth. Coverage is 0.900 at every horizon against a nominal 90%, and scaling the conformity score by the prediction gives locally adaptive intervals 22–48% narrower at the same coverage — which matters, because interval width varies by two orders of magnitude across counties. Two caveats we state rather than hide: cells within a county are not exchangeable, so this is a county-split check and not a formal guarantee; and the point forecast is unchanged by construction, so none of it moves the RMSE we are scored on. Interval calibration and point accuracy are separable here, and conformal prediction is why we can say so.

We do not claim separation from comparable submissions. Resampling 63 counties gives a score standard deviation of 0.0016 — roughly 45× the margin between our final model and our own earlier ones, and larger than the entire spread of the 2025 finalist field. A random 63-county draw moves our own score between 0.0055 and 0.0114. Any ranking at the fifth decimal is substantially a statement about which counties were held out.

5 Audit, limits, and prior work

Defects we found in our own measurements. A ledger of 246 contrasts is only as good as the harness that scored it, so we audited ours.

defect	consequence
bootstrap resampled <i>pooled</i> rather than per-horizon RMSE	every load-bearing comparison re-run; no verdict flipped
a production headline quoted the $t+1$ h column as the mean	had reversed one contingency decision; corrected, the shipped blend wins on both metrics
four builder-script names in the reproduction notes did not exist	corrected against <code>git ls-files</code> ; one file has no committed builder, now stated
a coverage table summed to 120 while claiming 129	missing row restored

Limits. The residual result is empirical, not information-theoretic: we show the per-county multiplier is unrecoverable by the features and learners we tested, not that it is unpredictable in principle. One member consumes external static

data; substituting a package-only variant costs 0.00004, measured, and ships as a one-line fallback. All numbers are county-holdout estimates on 239 training counties, not leaderboard scores.

Prior work. Operational outage modelling predicts storm totals or damage counts from forecast weather and does not produce hourly trajectories; that literature is dominated by tree ensembles on storm-level exposure features [13, 14], often with an explicit two-part structure for the zero mass [15]; now-casting work produces hourly forecasts at 1–6h horizons with the target’s own recent history available. Ours is neither: a 143-hour horizon, for counties whose history is withheld, with future weather known. The nearest general tools are the recent exogenous-covariate architectures — TimeXer [9], XLinear [10], TSMixer [11] — and covariate-aware pretrained forecasters such as Chronos-2 [12]. We ran all of them on this protocol; they appear in F1 and F2 above. What we did not find in prior work is the use of a residual budget as a filter applied *before* the experiments, with the full ledger — negative results included — reported as a test of it.

6 Conclusion and reproducibility

A pre-experiment residual diagnosis allocated attention across a closed set of intervention families; 210 controlled runs and 28 paired contrasts then tested those family-level predictions under county-held-out, nested validation. The model emerged from the interventions that survived, not from ranking configurations. The finding we would carry to another problem is that in a small-sample transfer regime a component’s marginal value is governed by the complementarity of its errors at least as much as by their size — which is why the three weakest models in our set are the three we could not remove.

`make_submission.py` rebuilds the prediction file end-to-end from the three provided CSVs with fixed seeds and no manual steps; a clean clone reproduces a byte-identical SHA-256. `canonical.py` regenerates every number quoted here from cached out-of-fold predictions, so any citation can be checked by re-running rather than by trusting. All of it is submitted as `code.zip` alongside this report, self-contained and needing no network access.

The same code, the full experiment catalogue in four for-

mats, and an interactive dashboard over it — a single self-contained page with all 210 runs sortable, the members plotted by accuracy against error correlation, and a builder that returns the exact score of any subset of 60 models you combine — are also at

<https://github.com/HamedKhosravi99/informs2026-model-catalogue>

The dashboard is served directly at <https://hamedkhosravi99.github.io/informs2026-model-catalogue/>

Both become reachable on 25 September 2026: the repository is private while the competition is open and is made public once the deadline has passed, so the committee can inspect it directly. Nothing there is needed to reproduce the submission — `code.zip` is complete on its own and needs no network access.

References

- [1] A. Zeng et al. Are transformers effective for time series forecasting? *AAAI*, 2023.
- [2] Y. Nie et al. A time series is worth 64 words: long-term forecasting with transformers. *ICLR*, 2023.
- [3] Y. Liu et al. iTransformer: inverted transformers are effective for time series forecasting. *ICLR*, 2024.
- [4] M. Tan et al. Are language models actually useful for time series forecasting? *NeurIPS*, 2024.
- [5] G. Huang et al. Snapshot ensembles: train 1, get M for free. *ICLR*, 2017.
- [6] H. Zhang et al. mixup: beyond empirical risk minimization. *ICLR*, 2018.
- [7] G. Cawley and N. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *JMLR*, 11:2079–2107, 2010.
- [8] T. Kim et al. Reversible instance normalization for accurate time-series forecasting against distribution shift. *ICLR*, 2022.
- [9] Y. Wang et al. TimeXer. *NeurIPS*, 2024.
- [10] Z. Ma et al. XLinear. arXiv:2601.09237, 2026.
- [11] S.-A. Chen et al. TSMixer. *TMLR*, 2023.
- [12] A. F. Ansari et al. Chronos-2. arXiv:2510.15821, 2025.
- [13] D. W. Wanik, E. N. Anagnostou, B. M. Hartman, M. E. B. Frediani, M. Astitha. Storm outage modeling for an electric distribution network in Northeastern USA. *Natural Hazards*, 2015.
- [14] Probabilistic and machine learning methods for uncertainty quantification in power outage prediction due to extreme events. *Nat. Hazards Earth Syst. Sci.* 23:1665, 2023.
- [15] Operational early warning of thunderstorm-driven power outages from open data: a two-stage machine learning approach. arXiv:2510.03959, 2025.